

## COURSE DESCRIPTION FORM

**INSTITUTION**

FAST-NUCES

**PROGRAM (S) TO BE EVALUATED** BS Computer Science/DS\_ (BDS-6C) (BCS-4C/4D)

### A. Course Description

|                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                   |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Course Code</b>                                                                                                       | AI-2002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>National Computing Education Accreditation</b> |
| <b>Course Title</b>                                                                                                      | Artificial Intelligence                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | NCEAC                                             |
| <b>Credit Hours</b>                                                                                                      | 3 + 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                   |
| <b>Prerequisites by Course(s) and Topics</b>                                                                             | Data Structures                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                   |
| <b>Assessment Instruments with Weights</b> (homework, quizzes, midterms, final, programming assignments, lab work, etc.) | <ol style="list-style-type: none"> <li>1. Assignments + Homework <b>(10 %)</b></li> <li>2. Project <b>(5 %)</b></li> <li>3. Quizzes <b>(10 %)</b></li> <li>4. Midterms <b>(30 %)</b></li> <li>5. Final Exam <b>(45 %)</b></li> </ol>                                                                                                                                                                                                                                                    |                                                   |
| <b>Course Instructor</b>                                                                                                 | Usman Anwer                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                   |
| <b>Office Address</b>                                                                                                    | F-23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                   |
| <b>Office Hours</b>                                                                                                      | <b>Tuesday-Thursday</b> 1:00 PM to 02:00 PM,                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                   |
| <b>Course TA</b>                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                   |
| <b>Email</b>                                                                                                             | usman.anwer@lhr.nu.edu.pk                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                   |
| <b>Textbook (or Laboratory Manual for Laboratory Courses)</b>                                                            | <ol style="list-style-type: none"> <li>1. Stuart Russell, Peter Norvig - Artificial Intelligence A Modern Approach - (3rd Edition)</li> <li>2. George F. Luger - Artificial Intelligence</li> </ol>                                                                                                                                                                                                                                                                                     |                                                   |
| <b>Reference Material</b>                                                                                                | <ol style="list-style-type: none"> <li>1. Tom Mitchell Machine Learning</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                      |                                                   |
| <b>Course Policies:</b>                                                                                                  | <ul style="list-style-type: none"> <li>• Quizzes may be announced or surprised. No Late Submissions or Makeup Quizzes.</li> <li>• Students bear all the responsibility for protecting their assignments. In case of cheating, both parties will be considered equally responsible.</li> <li>• Plagiarism in any work (labs, quizzes, Assignments, midterms, and final exams) from any source, Internet, or student will result in an F grade or deduction of absolute marks.</li> </ul> |                                                   |

| Course Goals<br>Mapping of CLOs & PLOs                                                                                    | <table><thead><tr><th></th><th>CLOs</th><th>PLOs</th></tr></thead><tbody><tr><td>1.</td><td>Summarize the notions of rational behavior and <b>intelligent agents</b>.</td><td>2</td></tr><tr><td>2.</td><td>Express appreciation of the goals, subareas, achievements, and difficulties of AI.</td><td>2</td></tr><tr><td>3.</td><td>Apply methods of <b>blind as well as informed search</b> on the problem related to the programs.</td><td>3</td></tr><tr><td>4.</td><td>Summarize concepts and approaches in knowledge representation, planning, learning, robotics, and other AI areas.</td><td>2</td></tr><tr><td>5.</td><td><b>Program AI applications</b> by implementing the concepts learnt in the course</td><td>4</td></tr></tbody></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      | CLOs                                                | PLOs | 1.   | Summarize the notions of rational behavior and <b>intelligent agents</b> . | 2 | 2. | Express appreciation of the goals, subareas, achievements, and difficulties of AI.                                                        | 2 | 3.  | Apply methods of <b>blind as well as informed search</b> on the problem related to the programs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3 | 4.  | Summarize concepts and approaches in knowledge representation, planning, learning, robotics, and other AI areas.                           | 2 | 5. | <b>Program AI applications</b> by implementing the concepts learnt in the course | 4 |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------|------|------|----------------------------------------------------------------------------|---|----|-------------------------------------------------------------------------------------------------------------------------------------------|---|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|--------------------------------------------------------------------------------------------------------------------------------------------|---|----|----------------------------------------------------------------------------------|---|
|                                                                                                                           | CLOs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | PLOs |                                                     |      |      |                                                                            |   |    |                                                                                                                                           |   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |     |                                                                                                                                            |   |    |                                                                                  |   |
| 1.                                                                                                                        | Summarize the notions of rational behavior and <b>intelligent agents</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2    |                                                     |      |      |                                                                            |   |    |                                                                                                                                           |   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |     |                                                                                                                                            |   |    |                                                                                  |   |
| 2.                                                                                                                        | Express appreciation of the goals, subareas, achievements, and difficulties of AI.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2    |                                                     |      |      |                                                                            |   |    |                                                                                                                                           |   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |     |                                                                                                                                            |   |    |                                                                                  |   |
| 3.                                                                                                                        | Apply methods of <b>blind as well as informed search</b> on the problem related to the programs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3    |                                                     |      |      |                                                                            |   |    |                                                                                                                                           |   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |     |                                                                                                                                            |   |    |                                                                                  |   |
| 4.                                                                                                                        | Summarize concepts and approaches in knowledge representation, planning, learning, robotics, and other AI areas.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2    |                                                     |      |      |                                                                            |   |    |                                                                                                                                           |   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |     |                                                                                                                                            |   |    |                                                                                  |   |
| 5.                                                                                                                        | <b>Program AI applications</b> by implementing the concepts learnt in the course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4    |                                                     |      |      |                                                                            |   |    |                                                                                                                                           |   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |     |                                                                                                                                            |   |    |                                                                                  |   |
| Topics Covered in the Course, with Number of Lectures on Each Topic<br>(assume 15-week instruction and one-hour lectures) | <table><thead><tr><th></th><th colspan="2">National Computing Education Accreditation<br/>NCEAC</th></tr><tr><th>Week</th><th>Lecture Details</th><th></th></tr></thead><tbody><tr><td>1</td><td><ul style="list-style-type: none"><li>– Introduction</li><li>– Agent and Agent Architectures</li><li>– Knowledge Representation</li></ul></td><td></td></tr><tr><td>2-5</td><td><p><b>PEAS</b></p><p><b>Strategies for State Space Search:</b></p><p><b>Blind/Uninformed/Brute-force Search</b></p><ul style="list-style-type: none"><li>1. Depth First Search (DFS)</li><li>2. Breadth First Search (BFS)</li><li>3. Iterative Deepening Search (IDS)</li><li>4. Uniform Cost Search (UCS)</li></ul><p>Direction of Search, Branching Factor</p><p><b>Heuristic/Informed Search</b></p><ul style="list-style-type: none"><li>1. Best First Search</li><li>2. A* Search</li></ul><p>Informedness, Monotonicity, Admissibility, Optimality, Completeness</p><p><b>Adversarial Search Algorithms (Game Playing)</b></p><ul style="list-style-type: none"><li>1. Minimax Search,<ul style="list-style-type: none"><li>a. Alpha-Beta Pruning,</li><li>b. Evaluation Functions</li></ul></li></ul><p><b>Local Search:</b></p><ul style="list-style-type: none"><li>1. Hill Climbing Search</li></ul><p><b>Evolutionary Search:</b> Genetic Algorithm</p><ul style="list-style-type: none"><li>1. Genetic Algorithm Introduction</li><li>2. Solution representation</li></ul></td><td></td></tr><tr><td>6-7</td><td><ul style="list-style-type: none"><li>3. Fitness evaluation function</li><li>4. Selection mechanism (Roulette wheel, Rank based,</li></ul></td><td></td></tr></tbody></table> |      | National Computing Education Accreditation<br>NCEAC |      | Week | Lecture Details                                                            |   | 1  | <ul style="list-style-type: none"><li>– Introduction</li><li>– Agent and Agent Architectures</li><li>– Knowledge Representation</li></ul> |   | 2-5 | <p><b>PEAS</b></p> <p><b>Strategies for State Space Search:</b></p> <p><b>Blind/Uninformed/Brute-force Search</b></p> <ul style="list-style-type: none"><li>1. Depth First Search (DFS)</li><li>2. Breadth First Search (BFS)</li><li>3. Iterative Deepening Search (IDS)</li><li>4. Uniform Cost Search (UCS)</li></ul> <p>Direction of Search, Branching Factor</p> <p><b>Heuristic/Informed Search</b></p> <ul style="list-style-type: none"><li>1. Best First Search</li><li>2. A* Search</li></ul> <p>Informedness, Monotonicity, Admissibility, Optimality, Completeness</p> <p><b>Adversarial Search Algorithms (Game Playing)</b></p> <ul style="list-style-type: none"><li>1. Minimax Search,<ul style="list-style-type: none"><li>a. Alpha-Beta Pruning,</li><li>b. Evaluation Functions</li></ul></li></ul> <p><b>Local Search:</b></p> <ul style="list-style-type: none"><li>1. Hill Climbing Search</li></ul> <p><b>Evolutionary Search:</b> Genetic Algorithm</p> <ul style="list-style-type: none"><li>1. Genetic Algorithm Introduction</li><li>2. Solution representation</li></ul> |   | 6-7 | <ul style="list-style-type: none"><li>3. Fitness evaluation function</li><li>4. Selection mechanism (Roulette wheel, Rank based,</li></ul> |   |    |                                                                                  |   |
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|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------|----------------------------------|
|                                                           | <p>Tournament)</p> <ol style="list-style-type: none"> <li>Genetic operators (Crossover, Mutation)</li> <li>Replacement method</li> <li>Generation gap, Elitism</li> </ol> <p><b>Supervised learning</b></p> <p><b>Regression</b></p> <ul style="list-style-type: none"> <li>Simple Linear Regression</li> </ul> <p><b>Machine Learning Introduction, Supervised Learning, Classification (ANN)</b></p> <p><b>8-10</b></p> <ul style="list-style-type: none"> <li>Architecture: Feed-Forward Neural Network</li> <li>Activation Functions (Linear, Sigmoid, Relu, tanh, Softmax)</li> <li>Training Algorithms: Perceptron and common activation functions, Linear separability of a perceptron</li> <li>Perceptron Learning Rule</li> <li>Multilayer Perceptron</li> </ul> <hr/> <p><b>11-12</b></p> <ul style="list-style-type: none"> <li>Back propagation algorithm (Gradient Descent Learning)</li> <li>Issues of ANN (Over-fitting, Under-fitting, Validation testing)</li> <li>Performance measures: Confusion Matrix (Binary &amp; Multi Class)</li> </ul> <hr/> <p><b>13</b></p> <p><b>Supervised learning</b></p> <ul style="list-style-type: none"> <li>Decision Trees</li> </ul> <p><b>Convolutional Neural Network (fwd prop only)</b></p> <p><b>Unsupervised learning - Clustering and its application</b></p> <p><b>14-15</b></p> <ul style="list-style-type: none"> <li>K-means algorithm,</li> <li>K-medoids algorithm</li> <li>Agglomerative clustering</li> </ul> |                         |                        |                                  |
| <b>Laboratory Projects/Experiments Done in the Course</b> | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |                        |                                  |
| <b>Programming Assignments Done in the Course</b>         | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |                        |                                  |
| <b>Class Time Spent on (in credit hours)</b>              | <b>Theory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Problem Analysis</b> | <b>Solution Design</b> | <b>Social and Ethical Issues</b> |
|                                                           | 30%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 30%                     | 35%                    | 5%                               |
| <b>Oral and Written Communications</b>                    | Every student must submit a Project (In Group) and appear for viva.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |                        |                                  |



**Instructor Name** \_\_\_\_\_  
**Instructor Signature** \_\_\_\_\_  
**Date** \_\_\_\_\_

**National Computing Education Accreditation Council**  
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